Processor Programming Reference (PPR) for AMD Family 19h Model 11h, Revision B2 Processors Volume 4 of 9

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6 System Management Unit (SMU)

6.1 SMU Overview

The System Management Unit (SMU) is a component of the SOC that is responsible for a variety of system and power management tasks during boot and runtime. The SMU uses two blocks, the AMD Secure Processor (ASP or MPASP; formerly PSP/MP0) and the Power Management microprocessor (MP1), in order to assist with many of these tasks.

6.2 Initialization

The SMU does not require software initialization. Upon successful boot-up the SMU is ready for operation.

6.3 SMU Features

This section describes SOC features controlled by the SMU.

6.3.1 Clocking

The processor contains multiple PLLs to create all necessary clock signals. The PMFW controls all clock frequencies based on current performance and power requirements. Software does not directly control clock frequencies.

6.3.2 Sideband Interface (SBI)

The sideband interface (SBI) supports the following interface(s):

- The Remote Management Interface (SB-RMI). Refer to the Advanced Platform Management Link (APML) Specification for details 1.2 [Reference Documents].
- The Temperature Sensor Interface (SB-TSI). Refer to the SBI Temperature Sensor Interface Specification for details 9 [SB Temperature Sensor Interface (SB-TSI)].

6.3.3 SMUIO and PWR

The SMUIO and PWR subcomponents provide interfaces to platform and on-chip power management blocks. SMUIO communicates with external voltage regulator through SVI3 interface.

The SVI3 topology is a 3-wire serial bus designed to support bi-directional data transfer between a single controller and multiple devices. The three wires consist of SVD (Serial VID Data), SVC (Serial VID Clock) and SVTI (Serial Telemetry VID Input).

SMUIO also provides a Golden Time Stamp Counter (TSC) to enable global time stamp counter synchronization.

6.3.4 Thermal (THM)

The THM subcomponent utilizes temperature sensors for temperature monitoring and reporting.

6.4 SMU Components

6.4.1 MP

6.4.1.1 AXI Direct Memory Access

6.4.1.1.1 MPDMA Registers

6.4.1.1.1.1 MPDMA Configuration Unit (CRU) Register

MPMPDMACRUx0[95...F4]10500 (MP::MPDMACRU::MPDMA_C2PMSG_0)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_0
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0500,0CF10500,09810500,09510500]; MPMPDMACRU=0000_0000h
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010500; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]10504 (MP::MPDMACRU::MPDMA_C2PMSG_1)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_0
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0504,0CF10504,09810504,09510504]; MPMPDMACRU=0000_0000h
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010504; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]10508 (MP::MPDMACRU::MPDMA_C2PMSG_2)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_0
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0508,0CF10508,09810508,09510508]; MPMPDMACRU=0000_0000h
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010508; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]1050C (MP::MPDMACRU::MPDMA_C2PMSG_3)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_0 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_050C,0CF1050C,0981050C,0951050C]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx0301050C; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]10510 (MP::MPDMACRU::MPDMA_C2PMSG_4)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_0 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0510,0CF10510,09810510,09510510]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010510; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]10514 (MP::MPDMACRU::MPDMA_C2PMSG_5)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_0 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0514,0CF10514,09810514,09510514]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010514; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]10518 (MP::MPDMACRU::MPDMA_C2PMSG_6)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_0 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0518,0CF10518,09810518,09510518]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010518; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]1051C (MP::MPDMACRU::MPDMA_C2PMSG_7)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_0 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_051C,0CF1051C,0981051C,0951051C]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx0301051C; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]10520 (MP::MPDMACRU::MPDMA_C2PMSG_8)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_0 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0520,0CF10520,09810520,09510520]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010520; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]10524 (MP::MPDMACRU::MPDMA_C2PMSG_9)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_0 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0524,0CF10524,09810524,09510524]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010524; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]10540 (MP::MPDMACRU::MPDMA_C2PMSG_16)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_1 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0540,0CF10540,09810540,09510540]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010540; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. message content

MPMPDMACRUx0[95...F4]10A90 (MP::MPDMACRU::MPDMA_C2PMSG_100)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_6
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0A90,0CF10A90,09810A90,09510A90]; MPMPDMACRU=0000_0000h

_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010A90; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10A94 (MP::MPDMACRU::MPDMA_C2PMSG_101)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_6
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0A94,0CF10A94,09810A94,09510A94]; MPMPDMACRU=0000_0000h

_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010A94; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10A98 (MP::MPDMACRU::MPDMA_C2PMSG_102)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_6
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0A98,0CF10A98,09810A98,09510A98]; MPMPDMACRU=0000_0000h

_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010A98; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10A9C (MP::MPDMACRU::MPDMA_C2PMSG_103)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_6
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0A9C,0CF10A9C,09810A9C,09510A9C]; MPMPDMACRU=0000_0000h

_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010A9C; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AA0 (MP::MPDMACRU::MPDMA_C2PMSG_104)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_6 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AA0,0CF10AA0,09810AA0,09510AA0]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AA0; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AA4 (MP::MPDMACRU::MPDMA_C2PMSG_105)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_6 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AA4,0CF10AA4,09810AA4,09510AA4]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AA4; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AA8 (MP::MPDMACRU::MPDMA_C2PMSG_106)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_6 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AA8,0CF10AA8,09810AA8,09510AA8]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AA8; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AAC (MP::MPDMACRU::MPDMA_C2PMSG_107)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_6 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AAC,0CF10AAC,09810AAC,09510AAC]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AAC; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AB0 (MP::MPDMACRU::MPDMA_C2PMSG_108)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_6
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AB0,0CF10AB0,09810AB0,09510AB0]; MPMPDMACRU=0000_0000h
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AB0; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AB4 (MP::MPDMACRU::MPDMA_C2PMSG_109)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_6
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AB4,0CF10AB4,09810AB4,09510AB4]; MPMPDMACRU=0000_0000h
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AB4; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AB8 (MP::MPDMACRU::MPDMA_C2PMSG_110)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_6
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AB8,0CF10AB8,09810AB8,09510AB8]; MPMPDMACRU=0000_0000h
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AB8; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10ABC (MP::MPDMACRU::MPDMA_C2PMSG_111)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_6
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0ABC,0CF10ABC,09810ABC,09510ABC]; MPMPDMACRU=0000_0000h
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010ABC; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AC0 (MP::MPDMACRU::MPDMA_C2PMSG_112)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AC0,0CF10AC0,09810AC0,09510AC0]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AC0; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AC4 (MP::MPDMACRU::MPDMA_C2PMSG_113)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AC4,0CF10AC4,09810AC4,09510AC4]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AC4; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AC8 (MP::MPDMACRU::MPDMA_C2PMSG_114)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AC8,0CF10AC8,09810AC8,09510AC8]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AC8; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10ACC (MP::MPDMACRU::MPDMA_C2PMSG_115)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0ACC,0CF10ACC,09810ACC,09510ACC]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010ACC; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AD0 (MP::MPDMACRU::MPDMA_C2PMSG_116)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AD0,0CF10AD0,09810AD0,09510AD0]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AD0; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AD4 (MP::MPDMACRU::MPDMA_C2PMSG_117)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AD4,0CF10AD4,09810AD4,09510AD4]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AD4; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AD8 (MP::MPDMACRU::MPDMA_C2PMSG_118)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AD8,0CF10AD8,09810AD8,09510AD8]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AD8; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10ADC (MP::MPDMACRU::MPDMA_C2PMSG_119)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0ADC,0CF10ADC,09810ADC,09510ADC]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010ADC; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AE0 (MP::MPDMACRU::MPDMA_C2PMSG_120)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_7
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AE0,0CF10AE0,09810AE0,09510AE0]; MPMPDMACRU=0000_0000h

_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AE0; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AE4 (MP::MPDMACRU::MPDMA_C2PMSG_121)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_7
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AE4,0CF10AE4,09810AE4,09510AE4]; MPMPDMACRU=0000_0000h

_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AE4; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AE8 (MP::MPDMACRU::MPDMA_C2PMSG_122)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_7
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AE8,0CF10AE8,09810AE8,09510AE8]; MPMPDMACRU=0000_0000h

_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AE8; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AEC (MP::MPDMACRU::MPDMA_C2PMSG_123)

Read-write. Reset: 0000_0000h.

Inbound mailbox register (external->MPDMA)
External access control by C2PMSG_ATTR_7
Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable

_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AEC,0CF10AEC,09810AEC,09510AEC]; MPMPDMACRU=0000_0000h

_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AEC; MPDMAAXIXBAR=0000_0000h

Bits	Description
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AF0 (MP::MPDMACRU::MPDMA_C2PMSG_124)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AF0,0CF10AF0,09810AF0,09510AF0]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AF0; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AF4 (MP::MPDMACRU::MPDMA_C2PMSG_125)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AF4,0CF10AF4,09810AF4,09510AF4]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AF4; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AF8 (MP::MPDMACRU::MPDMA_C2PMSG_126)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AF8,0CF10AF8,09810AF8,09510AF8]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AF8; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

MPMPDMACRUx0[95...F4]10AFC (MP::MPDMACRU::MPDMA_C2PMSG_127)	
Read-write. Reset: 0000_0000h.	
Inbound mailbox register (external->MPDMA) External access control by C2PMSG_ATTR_7 Reset by Cpl_VDDCR_<dom>_Resetn (0-> RESET, 1 -> out of RESET), where <dom>=VDD_SOC, VDD_S5, etc. as applicable	
_instMPDMA1_dup[3:0,5:4]_aliasSMN; MPMPDMACRUx[0[F4,ED,E6]1_0AFC,0CF10AFC,09810AFC,09510AFC]; MPMPDMACRU=0000_0000h	
_instMPDMA_aliasMPDMA; MPDMAAXIXBARx03010AFC; MPDMAAXIXBAR=0000_0000h	
Bits	**Description**
31:0	**CONTENT**. Read-write. Reset: 0000_0000h. CPU to MPDMA message (32-bit register)

6.4.2 FUSE

6.4.2.1 Registers

Table 148: SMUFUSEx000006CC [MaxPPT[4:0]]

Read-only.

_aliasSMN; SMUFUSEx000006CC; SMUFUSE=0005_D000h	
Bits	**Description**
31:27	MaxPPT[4:0]
26:17	PPT[9:0]
16:0	Reserved

Table 149: SMUFUSEx000006D0 [MaxTDP[9:0],MaxPPT[9:5]]

Read-only.	
_aliasSMN; SMUFUSEx000006D0; SMUFUSE=0005_D000h	
Bits	**Description**
31:25	Reserved
24:15	MaxTDP[9:0]
14:5	Reserved
4:0	MaxPPT[9:5]

6.4.3 Thermal (THM)

The thermal block contains all the features related to temperature sensing, control, and reporting. It includes:

- Temperature collection and calculation logic.
- Fan speed control for off-chip fans.
- Temperature reporting through the APML interface.

6.4.3.1 Registers

SMUTHMx00000000 (SMU::THM::THM_TCON_CUR_TMP)	
Reset: 0000_0000h.	
_aliasSMN; SMUTHMx00000000; SMUTHM=0005_9800h	
Bits	**Description**
31:21	**CUR_TEMP**. Read-only. Reset: 000h. Provides current control temperature (T ctl) after the slew-rate controls have been applied.
20	Reserved.
19	**CUR_TEMP_RANGE_SEL**. Read-write. Reset: 0. 0=Report on 0C to 225C scale range. 1=Report on -49C to 206C scale range.
18:0	Reserved.

SMUTHMx00000004 (SMU::THM::THM_TCON_HTC)	
Reset: 0000_0000h.	
_aliasSMN; SMUTHMx00000004; SMUTHM=0005_9800h	
Bits	**Description**
31:16	Reserved.
15	**DIS_PROCHOT_PIN_IN**. Read-write. Reset: 0. Disable external Prochot to impact internal logic.
14:4	Reserved.
3	**INTERNAL_PROCHOT**. Read-only. Reset: 0. PROCHOT triggered by internal temperature high.
2	**EXTERNAL_PROCHOT**. Read-only. Reset: 0. PROCHOT triggered by external PROCHOT pad being asserted.
1:0	Reserved.

SMUTHMx00000380 (SMU::THM::SMUSBI_SBIREGADDR)	
Read-write. Reset: 0000_0000h.	
_aliasSMN; SMUTHMx00000380; SMUTHM=0005_9800h	
Bits	**Description**
31:11	Reserved.
10:9	**SIZE**. Read-write. Reset: 0h. N+1 bytes to be read or written
8	**TSI_RMI_SEL**. Read-write. Reset: 0. When set to 0 selects SB-TSI register set; otherwise selects SB-RMI registers.
7:0	**Address**. Read-write. Reset: 00h. Select SBI internal register address for Read or Write.

SMUTHMx00000384 (SMU::THM::SMUSBI_SBIREGDATA)	
Read-write. Reset: 0000_0000h.	
_aliasSMN; SMUTHMx00000384; SMUTHM=0005_9800h	
Bits	**Description**
31:0	**SBI_REGDATA**. Read-write. Reset: 0000_0000h. Read or Write data for SBI internal register address.

SMUTHMx00000394 (SMU::THM::SMUSBI_ERRATA_STAT_REG)	
Read-only. Reset: 0000_0000h.	
_aliasSMN; SMUTHMx00000394; SMUTHM=0005_9800h	
Bits	**Description**
31:0	**ERRATA_STAT_REG**. Read-only. Reset: 0000_0000h. Errata status.